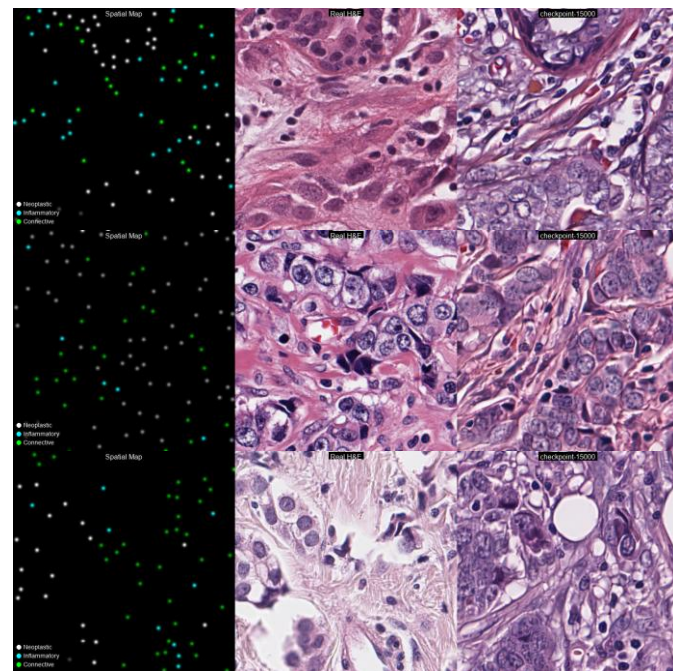
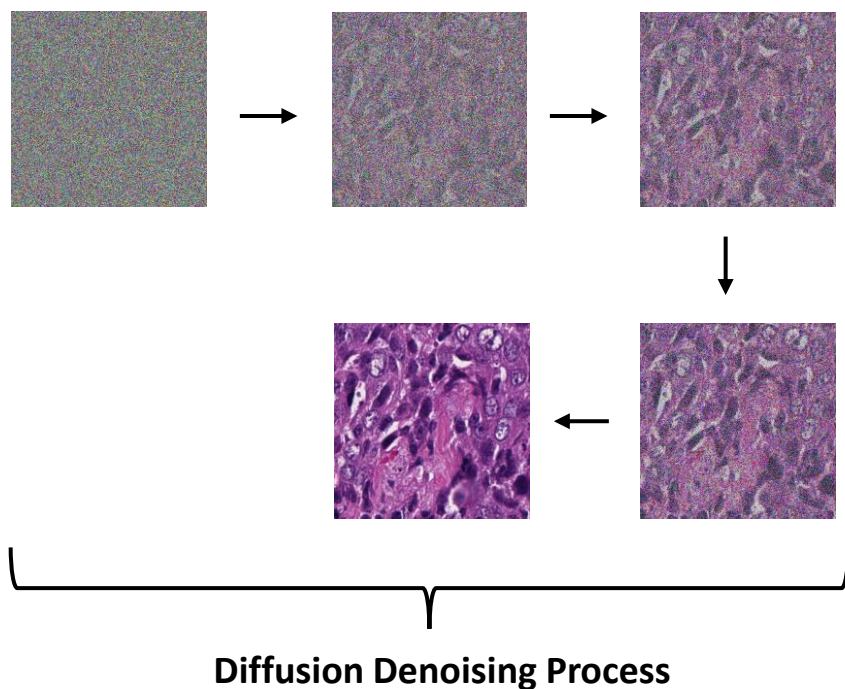


CPathoGen: A Framework for Controllable Histopathology

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Abstract: We created CPathoGen, a controllable AI image generator designed to create realistic digital snapshots of cancer tissue. While typical AI tools can be unpredictable, CPathoGen acts like a precision-guided artist because you can tell it exactly where to place specific cells and how aggressive a tumor should look. Our goal is to use these custom-made images to stress-test other medical AI models. By creating what-if scenarios, such as moving a few cells or changing a tissue's texture, we can identify exactly where diagnostic software might fail. This ensures that the technology used to detect cancer is as reliable and robust as possible.



Conclusion: This study attempts to build a framework to generatively interpret foundational models by enabling researchers to produce counterfactual Hematoxylin and Eosin (H&E) stained digital tiles. In future we aim to build upon this framework to further enhance the image quality and reducing the FID to accepted research standards.